

Infrastructure Governance Report

Report ID: a1b94285 Generated: 2026-07-20T09:57:20.062517

Files: azure-average.tf

Score Summary

Dimension	Score / 100
Overall	76.6
Security	65.0
Reliability	83.0
Cost	93.0
Architecture	70.0

Compliance Posture

Framework	Version	Score	Passed	Failed
CIS Azure Foundations Benchmark	3.0	66.7%	8	4
NIST 800-53 Rev 5	5	77.8%	14	4

CIS Azure Foundations Benchmark — Control Detail

Status	Control	Description
FAIL	CIS-Azure-6.2	Ensure that NSGs do not allow inbound RDP from the internet
FAIL	CIS-Azure-6.3	Ensure that NSGs do not allow inbound SSH from the internet
FAIL	CIS-Azure-8.4	Ensure that the soft delete is enabled and retention period is set on Key Vaults
FAIL	CIS-Azure-8.5	Ensure that the purge protection is enabled on Key Vaults
PASS	CIS-Azure-3.1	Ensure that 'Secure transfer required' is set to 'Enabled' on storage accounts
PASS	CIS-Azure-3.15	Ensure the minimum TLS version for storage accounts is set to Version 1.2
PASS	CIS-Azure-3.7	Ensure that 'Public access level' is disallowed for storage accounts
PASS	CIS-Azure-3.8	Ensure default network access rule for storage accounts is set to deny
PASS	CIS-Azure-4.3.5	Ensure no SQL server firewall rule allows ingress from 0.0.0.0/0 to 255.255.255.255
PASS	CIS-Azure-7.3	Ensure that managed disks are encrypted with customer-managed keys
PASS	CIS-Azure-8.7	Ensure that Azure RBAC (Role-Based Access Control) is used to control access to AKS resources
PASS	CIS-Azure-9.2	Ensure that App Service is configured with HTTPS-only access

NIST 800-53 Rev 5 — Control Detail

Status	Control	Description
FAIL	NIST-CP-7	Alternate Processing Site
FAIL	NIST-CP-9	System Backup

FAIL	NIST-SC-28	Protection of Information at Rest
FAIL	NIST-SC-7	Boundary Protection
PASS	NIST-AC-2	Account Management
PASS	NIST-AC-3	Access Enforcement
PASS	NIST-AC-6	Least Privilege
PASS	NIST-AU-12	Audit Record Generation
PASS	NIST-AU-2	Event Logging
PASS	NIST-CM-3	Configuration Change Control
PASS	NIST-CP-10	System Recovery and Reconstitution
PASS	NIST-IA-5	Authenticator Management
PASS	NIST-MA-2	Controlled Maintenance
PASS	NIST-SC-5	Denial of Service Protection
PASS	NIST-SC-6	Resource Availability
PASS	NIST-SC-8	Transmission Confidentiality and Integrity
PASS	NIST-SI-11	Error Handling
PASS	NIST-SI-4	System Monitoring

Findings Appendix

Security Agent — Score 65.0/100 — 4 findings

[CRITICAL] Azure NSG rule open to internet

Resource: azurerm_network_security_rule.allow_ssh

azurerm_network_security_rule.allow_ssh allows inbound from * on port 22.

Recommendation: Restrict source_address_prefix to specific CIDR blocks.

Controls: CIS-Azure-6.2, CIS-Azure-6.3, NIST-SC-7

[MEDIUM] Key Vault without purge protection

Resource: azurerm_key_vault.main

azurerm_key_vault.main can be permanently deleted. Secrets/keys may be irrecoverably lost.

Recommendation: Set purge_protection_enabled = true.

Controls: CIS-Azure-8.5, NIST-CP-9, NIST-SC-28

[MEDIUM] Key Vault without soft delete retention

Resource: azurerm_key_vault.main

azurerm_key_vault.main has no soft_delete_retention_days configured.

Recommendation: Set soft_delete_retention_days (e.g., 90) for recoverability.

Controls: CIS-Azure-8.4, NIST-CP-9

[MEDIUM] Non-redundant SQL Database Configuration

Resource: azurerm_mssql_database.main

The MSSQL database is configured with 'zone_redundant = false'. This means the database and its associated data are not protected by zone redundancy, increasing the risk of downtime or data loss if a single availability zone fails.

Recommendation: Set 'zone_redundant' to true for critical databases to ensure high availability and disaster recovery across multiple zones.

Reliability Agent — Score 83.0/100 — 4 findings

[MEDIUM] Azure SQL database not zone-redundant

Resource: azurerm_mssql_database.main

azurerm_mssql_database.main is not zone-redundant. Single zone failure risk.

Recommendation: Set zone_redundant = true for production databases.

Controls: NIST-CP-7

[MEDIUM] VM without availability zone/set

Resource: azurerm_linux_virtual_machine.app

azurerm_linux_virtual_machine.app has no availability set or zone specified.

Recommendation: Assign to an availability set or specify an availability zone.

Controls: NIST-CP-7

[MEDIUM] Key Vault Lacks Deletion Protection

Resource: azurerm_key_vault.main

The Azure Key Vault resource has `purge\_protection\_enabled = false`. If this vault is accidentally deleted, sensitive secrets and keys will be permanently lost without any recovery mechanism.

Recommendation: Set `purge\_protection\_enabled = true` to enable soft-delete protection for the Key Vault.

[LOW] Azure SQL without inline long-term retention

Resource: azurerm_mssql_database.main

azurerm_mssql_database.main has no inline long_term_retention_policy. This may be configured via a separate resource (azurerm_mssql_database_extended_auditing_policy).

Recommendation: Configure long_term_retention_policy inline or via a separate retention policy resource.

Controls: NIST-CP-9

Cost Agent — Score 93.0/100 — 3 findings

[MEDIUM] SQL Database Zone Redundancy Disabled

Resource: azurerm_mssql_database.main

The Azure SQL database is configured with `zone\_redundant = false`. This configuration sacrifices high availability and disaster recovery capabilities, which can lead to higher operational risk and potential costs if failover mechanisms are required.

Recommendation: Enable zone redundancy for the SQL database to ensure resilience. Review the cost implications of enabling zone redundancy versus the current setup.

[LOW] Missing Storage Lifecycle Policies

Resource: azurerm_storage_account.data

The storage account (`appdatastorage123`) and associated disks do not have defined lifecycle management policies. This means data retention, archival, or tiering opportunities (e.g., moving older data to cooler tiers) are missed, potentially leading to unnecessary storage costs.

Recommendation: Implement Azure Storage Lifecycle Management policies on the storage account and associated disks to automatically transition data to cooler storage tiers based on age.

[INFO] VM Instance Sizing Review Needed

Resource: azurerm_linux_virtual_machine.app

The Linux VM is provisioned using `Standard\_B2ms`. While this is a specific size, without workload metrics (CPU/Memory utilization), it is impossible to confirm if this instance type represents optimal cost efficiency. It requires validation against actual usage.

Recommendation: Conduct a thorough review of the VM's historical CPU and memory utilization metrics to determine if the `Standard\_B2ms` size is right-sized, or if a smaller/different family instance type can achieve the same performance at a lower cost.

Compliance Agent — Score 72.2/100 — 2 findings

[MEDIUM] CIS Azure Foundations Benchmark: 4 control(s) failing

Resource: cis_azure

CIS Azure Foundations Benchmark (v3.0) compliance is 66.7% — failing controls: CIS-Azure-6.2, CIS-Azure-6.3, CIS-Azure-8.4, CIS-Azure-8.5.

Recommendation: Remediate the findings mapped to these controls to raise the CIS Azure Foundations Benchmark compliance score.

[MEDIUM] NIST 800-53 Rev 5: 4 control(s) failing

Resource: nist_800_53

NIST 800-53 Rev 5 (v5) compliance is 77.8% — failing controls: NIST-CP-7, NIST-CP-9, NIST-SC-28, NIST-SC-7.

Recommendation: Remediate the findings mapped to these controls to raise the NIST 800-53 Rev 5 compliance score.

Resilience Agent — Score 100.0/100 — 2 findings

[HIGH] Single point of failure: azurerm_resource_group.main

Resource: azurerm_resource_group.main

azurerm_resource_group.main (azurerm_resource_group) is a single point of failure (high-fan-in and articulation-point). 11 resource(s) depend on it: azurerm_key_vault.main, azurerm_linux_virtual_machine.app, azurerm_mssql_database.main, azurerm_mssql_server.main,

azurerem_network_interface.app (+6 more). Removing it would partition the architecture into disconnected components.

Recommendation: Add redundancy for this resource (e.g. replicas / Multi-AZ / a standby or read replica) or decouple dependents so its loss does not cascade across the system.

[LOW] Single point of failure: azurerem_mssql_server.main

Resource: azurerem_mssql_server.main

azurerem_mssql_server.main (azurerem_mssql_server) is a single point of failure (articulation-point). 1 resource(s) depend on it: azurerem_mssql_database.main. Removing it would partition the architecture into disconnected components.

Recommendation: Add redundancy for this resource (e.g. replicas / Multi-AZ / a standby or read replica) or decouple dependents so its loss does not cascade across the system.